

Information Retrieval

section (2)

Stemming

- Stemming is the process of producing morphological variants of a root/base word.
- **Stemming programs** are commonly referred to **as stemming algorithms or stemmers**.
- A stemming algorithm reduces the words “retrieval”, “retrieved”, “retrieves” reduce to the stem “retrieve”.
- Removal of affixes (usually suffixes) to arrive at a base form

Stemming

- The process of stemming is used to normalize text and make it easier to process.
- It is an important step in text pre-processing, and it is commonly used in information retrieval and text mining applications.

for example compressed and
compression are both accepted
as equivalent to compress.



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Errors in Stemming

- There are mainly two errors in stemming
 - Over-stemming and Under-stemming.

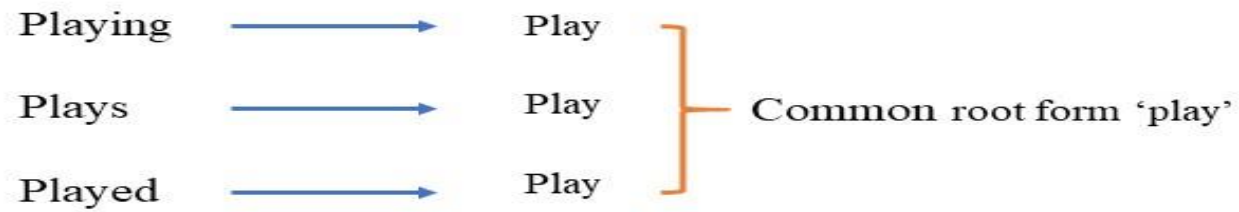
1. Over-stemming

- Removing too many characters, leading to the loss of important information.
- is not the correct root form of a word.
 - “beautiful” → “beauti”
 - “happiest” →” happi”

2. Under-stemming

- stemming algorithm fails to reduce a word to its root form
- Children →” children” correct is →“child”
- Happening→” happenin” correct is → “happen”

Stemming Example



am, are, is → be

Car cars, car's, cars' → car

Using above mapping a sentence could be normalized as follows:

the boy's cars are different colors → the boy car be differ color

Stemming

- For the English language, you can choose between
 - **Nltk**
 - PorterStemmer
 - SnowballStemmer
 - LancasterStemmer
 - WordNetLemmatizer
 - **Spacy** → lemma

PorterStemmer

- *PorterStemmer is known for its simplicity and speed.* It is commonly useful in **Information Retrieval Environments** known as **IR Environments** for fast recall and fetching of search queries
- **Words having the same stem will have a similar meaning. For example,**

CONNECT

CONNECTIONS-----> CONNECT

CONNECTED-----> CONNECT

CONNECTING-----> CONNECT

CONNECTION-----> CONNECT

SnowBallStemmer

- It is a stemming algorithm which is also known as the Porter2 stemming algorithm as it is a better version of the Porter Stemmer since some issues of it were fixed in this stemmer.
- Snowball Stemmer is **more aggressive** than Porter Stemmer.
- Some issues in Porter Stemmer were fixed in Snowball Stemmer.

LanCasterEMMER

- LancasterStemmer is simple, but heavy stemming due to iterations and over-stemming may occur.
- Over-stemming causes the stems to be not linguistic, or they may have no meaning.
- LancasterStemmer produces an even shorter stem than porter because of iterations and over-stemming is occurred.

lemmatization

- Lemmatization is the process of reducing a word to its lemma, or dictionary/base form.
- The goal is to group all the inflected forms of a word so they can be treated as a single item.

cats	→	cat
dishes		dish
wolves		wolf
are		be
stopping		stop
enjoyed		enjoy

Stemming vs lemmatization

Aspect	Stemming	Lemmatization
Technique type	Text normalization	Text normalization
Process	Reduces a word to its root	Reduces a word to its base form (lemma)
Example	running → run	mice → mouse
Consideration	Does not consider part of speech	Considers part of speech (POS)
Goal	Reduce variation in word forms and improve matching accuracy	Reduce variation in word forms and improve matching accuracy

Stemming vs lemmatization

- Stemming

- produced by “**stemmers**”
- produces a word’s “stem”

- am → am

- the going → the go

- having → hav

- fast and simple (pattern-based)

- Lemmatization

- produced by “**lemmatizers**”
- produces a word’s “lemma”

- am → be

- the going → the going

- having → have

- requires: a dictionary and PoS